

A Simple Observer for Nonlinear Systems in x-coordinate using Lipschitz Extension Technique

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Abstract—In this paper, we present new method for observer construction of nonlinear system - Lipschitz extension technique. Using the lipschitz extension technique, we can relieve global lipschitz assumption. We consider the problem of constructing x-coordinate observer. In general, proposed technique need inverse of state transformation to construct x-coordinate observer. We solved this problem with specific Lipschitz extension technique in rectangular region which does not require inverse of the transformation.

I. INTRODUCTION

The observer problem is to estimate asymptotically the state of a given plant with the input and output information of the plant. Contrary to linear systems, the problem has not yet been wholly solved in the general sense and there are several particular classes of nonlinear systems (A.J.Krener and Isidori, 1983; A.J.Krener and Respondek, 1985; Walcott and Zak, 1987; Tsinias, 1989; Gauthier *et al.*, 1992; Dawson *et al.*, 1992; Gauthier and Kupka, 1994; Besancon and Hammouri, 1996; Hou and Pugh, 1999; Shim *et al.*, 1999), for which the observer exists.

Roughly speaking, there are two major approaches to the state observer design for nonlinear systems; they are the *observer canonical form* approach and the *observable canonical form* approach, according to the terminology of Keller (1987). In the former case, a given nonlinear system is transformed into the following form:

$$\dot{x} = Ax + \gamma(*), \quad y = Cx \quad (1)$$

where (A, C) is observable (or detectable) pair, and $\gamma(*)$ is a vector field of which the arguments $(*)$ are the known quantities, like input and output, or sometimes their derivatives. Then, the observer construction becomes straightforward with L such that $(A - LC)$ is Hurwitz, i.e., $\hat{\dot{x}} = A\hat{x} + \gamma(*) - L(C\hat{x} - y)$ is the observer. This is the approach of *Linearized Error Dynamics* prevalent in the literature (Bestle and Zeitz, 1983; Keller, 1987; A.J.Krener and Isidori, 1983; A.J.Krener and Respondek, 1985; Birk and Zeitz, 1988; Xia and Gao, 1988; Xia and Gao, 1989; Li and Tao, 1986; Proychev and Mishkov, 1993; Hou and Pugh, 1999). However, these designs have a bottleneck that one should solve some partial differential equations to find the suitable transformation.

On the Contrary, the latter approach has the benefit that the change of coordinates is relatively easy because the transformation is intrinsically based on the mapping $[h(x), L_f h(x), \dots, L_f^{n-1} h(x)]^T$ which constitutes the *observation space* (Nijmeijer and van der Schaft, 1990). Moreover, the observable canonical form is directly related to the notion of observability, which results in the conceptual coincidence. In this paper, we begin by the *uniform observability* raised by Gauthier and G.Bornard (1981) and

present a novel design method for some class of nonlinear systems.

II. PRELIMINARIES AND BASIC RESULTS

A. Lipschitz function

In this subsection, some useful facts are presented that are used in developing the *Lipschitz extension technique*.

Definition 1. A function is said to be Lipschitz when it satisfies the inequality

$$\|f(x) - f(y)\| \leq L\|x - y\|, \quad x, y \in \mathbb{R}^n \quad (2)$$

where L is called Lipschitz constant.

The following lemma is trivial by continuity.

Lemma 1. If a function is Lipschitz on a set A and continuous on $A \cup \partial A$, it is also Lipschitz on the closure $A \cup \partial A$ with the same Lipschitz constant.

Lemma 2. Product of two Lipschitz functions on a compact set is also Lipschitz on the set.

Proof. Let f and g are Lipschitz on a compact set A . Then,

$$\begin{aligned} \|f(x)g(x) - f(y)g(y)\| & \leq \|f(x)g(x) - f(y)g(x)\| + \|f(y)g(x) - f(y)g(y)\| \\ & \leq L_f \|g(x)\| \|x - y\| + L_g \|f(y)\| \|x - y\| \\ & \leq L \|x - y\| \end{aligned}$$

where L is the maximum value of $L_f \|g(x)\|$ and $L_g \|f(x)\|$ on the compact set A . $\square$

Lemma 3. Suppose two sets $A \subset \mathbb{R}^n$ and $B \subset \mathbb{R}^n$ are disjoint and $A \cup B$ is convex. When a function f is continuous on $\mathbb{R}^n$ and Lipschitz on each set, then f is also Lipschitz on $A \cup B$.

Proof. Choose $x \in A$ and $y \in B$ and $z \in \partial A$, where z is on the line connecting x and y . Such z exists since $A \cup B$ is convex. Then,

$$\begin{aligned} \|f(x) - f(y)\| & \leq \|f(x) - f(z)\| + \|f(z) - f(y)\| \\ & \leq L_A \|x - z\| + L_B \|z - y\| \\ & \leq L(\|x - z\| + \|z - y\|) = L\|x - y\| \end{aligned}$$

where L_A and L_B are Lipschitz constants on A and B , respectively, and $L = \max\{L_A, L_B\}$. $\square$

B. Definition of Uniformly Observability

The observability for nonlinear systems is defined as the existence of a specific input $u(t)$ for any two different initial states, such that the states are distinguishable by the input and the output of the system (Nijmeijer and van der Schaft, 1990). Thus, even for an observable system, it is possible for two states to be indistinguishable by an input other than the specific $u(t)$. (See, for example, Vidyasagar (1993, p. 415).) This is not the case of linear systems in which the input u is multiplied by a constant vector field.

Yet, there is another notion of observability for nonlinear systems, which is a bit stronger than usual observability concept. Further discussions for the uniform observability are well organized in Gauthier and G.Bornard (1981).

Definition 2. A system is uniformly observable when it is observable for any input. More specifically, for any input $u(t)$ and any finite time interval $[0, T]$, $T > 0$, such that the trajectory of the system exists, the output $y(t)$, $t \in [0, T]$, determines the initial states $x(0)$.

C. Gauthier's High Gain Observer

The uniform observability in the previous subsection is closely related to the existence of high gain state observer proposed by Gauthier *et al.* (1992). That is, if a system is uniformly observable, then an observer always exists for the system under some technical assumptions. The design of the observer has been presented for input affine systems in (Gauthier *et al.*, 1992) and for input non-affine systems in (Gauthier and Kupka, 1994), respectively. Consider the single-output and input non-affine system:

$$\begin{aligned} \dot{z} &= a(z) + b(z, u) \\ y &= z_1 \end{aligned} \quad (3)$$

$$= \begin{bmatrix} z_2 \\ z_3 \\ \vdots \\ z_n \\ a_n(z) \end{bmatrix} + \begin{bmatrix} b_1(z_1, u) \\ b_2(z_1, z_2, u) \\ \vdots \\ b_{n-1}(z_1, \dots, z_{n-1}, u) \\ b_n(z_1, \dots, z_n, u) \end{bmatrix}$$

where $b(z, 0) = 0$.

Assumption 1. $a_n(z)$ is globally Lipschitz in z , and $b_i(z, u)$ is globally Lipschitz in z uniformly in u .

A sufficient conditions for the assumption that $b_i(z, u)$ is Lipschitz in z uniformly in u , is that $b_i(z, u)$ is Lipschitz in z and at the same time the input u is bounded. Therefore, we can assume the alternative:

Assumption 2. $a_n(z)$ and $b_i(z, u)$ are globally Lipschitz in z , and the input u is bounded.

Here, the value of the Lipschitz coefficient is not restricted and can be arbitrarily large. Then, the Gauthier's observer is given by

$$\begin{aligned} \dot{\hat{z}} &= a(\hat{z}) + b(\hat{z}, u) - G^{-1}C^T(C\hat{z} - y) \\ &= \begin{bmatrix} \hat{z}_2 \\ \hat{z}_3 \\ \vdots \\ \hat{z}_n \\ a_n(\hat{z}) \end{bmatrix} + \begin{bmatrix} b_1(\hat{z}_1, u) \\ b_2(\hat{z}_1, \hat{z}_2, u) \\ \vdots \\ b_{n-1}(\hat{z}_1, \dots, \hat{z}_{n-1}, u) \\ b_n(\hat{z}_1, \dots, \hat{z}_n, u) \end{bmatrix} - G^{-1}C^T(C\hat{z} - y) \end{aligned} \quad (4)$$

Where $\hat{z}$ is the estimated state of z and G is the unique positive definite solution of

$$0 = -\theta G - A^T G - G A + C^T C \quad (5)$$

where,

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{bmatrix}, \quad C = [1 \ 0 \ 0 \ 0 \ \dots \ 0]$$

and the positive constant θ is to be chosen in the following lemma.

Lemma 4 (Exponential Observer (Gauthier *et al.*, 1992)). Consider the plant and the observer given by (3) and (5). Under Assumption 1 or 2, there exists a constant $\theta_1^* \geq 1$ such that for any $\theta > \theta_1^*$, the observer (5) guarantees

$$\|\hat{z}(t) - z(t)\| \leq K(\theta) \exp\left(-\frac{\theta}{4}t\right) \|\hat{z}(0) - z(0)\|. \quad (6)$$

for $t \geq 0$ with some function $K(\theta)$.

Remark 1. Contrary to the linearized error dynamics approach which is based on the exact cancellation of nonlinear terms, Gauthier's observer suppress some nonlinearity by the high injection gain. However, it is different from the approach of (Rajamani and Cho, 1995; Rayamani, 1998) in which the growth of nonlinearity is restricted. In that approach, the observer gain is selected independently of the Lipschitz coefficient of the nonlinearity. Therefore, one should verify the convergence of the error after assigning the gain and it may happen that the appropriate gain cannot be found. On the other hand, in the Gauthier's approach, the gain is chosen by the knowledge of the Lipschitz coefficient of nonlinearity. Thus, once the nonlinearity is known to be Lipschitz, there always exists a suitable gain. See Gauthier *et al.* (1992) or Gauthier and Kupka (1994) to obtain theta, and more specific proof of lemma 4.

III. LIPSCHITZ EXTENSION TECHNIQUE

A. Lipschitz Extension

Although the Gauthier's observer is a good solution for the observer problem of uniformly observable systems, Assumption 1 (or 2) is the main restriction of the approach. In order to overcome the restriction, we present the *Lipschitz extension* technique in this section. For the technique to be applicable, we assume the boundedness of the trajectory $z(t)$ of the plant:

$$\dot{z} = a(z) + b(z, u) \quad (7)$$

Assumption 3. There is a compact set $Z \subset \mathbb{R}^n$ such that the trajectory $z(t)$ of (8) is contained in Z for all $t \geq 0$.

This is a sacrifice for relieving the Lipschitz property of the vector fields a and b . Now since $z(t)$ is confined to the bounded set Z , the dynamics (8) can be modified outside the region Z . That is, when the system (8) satisfies Assumption 3, it doesn't make any difference that we regard the system model, instead of (8), as

$$\dot{z} = \hat{a}(z) + \hat{b}(z, u) \quad (8)$$

where $\hat{a}$ and $\hat{b}$ are modified outside Z to be globally Lipschitz in z . Then, since $\hat{a}$ and $\hat{b}$ are Lipschitz in z , the system (9) (equivalently (8)) always admits the Gauthier's observer when the input u is bounded.

There is another advantage of the extension. Suppose that the system model (8) is only valid on a bounded region Z , in particular, that the model (8) has a singularity outside Z . This is often the case because the modeling is usually done in the operating region. For example, the usual modeling process for magnetic levitation system gives singularity when the magnet is attached to the iron deck. (See Joo *et al.* (1994)) Now suppose there is an observer which traces the true states asymptotically. Even though the estimate converges to the true state, it may overshoot at the initial stage. That is, it may go through the region outside Z where the vector fields a or b may not be well defined. Therefore, the modification of the system model outside the region Z is also a good solution to this accident. As a result, the observer uses the modified model to estimate the true state.

From now on, we precisely define this modification as the *Lipschitz extension*, and show the practical methods for the extension.

Definition 3 (Lipschitz Extension). For a C^1 function $f(x, u) : \bar{X}(\subset \mathbb{R}^n) \times \mathbb{R}^p \rightarrow \mathbb{R}^m$ and a compact set $X \subset \bar{X}$, a function $\hat{f}(x, u) : \mathbb{R}^n \times \mathbb{R}^p \rightarrow \mathbb{R}^m$ is called Lipschitz extension of $f(x, u)$ from X if

- (i) $\hat{f}(x, u)$ is continuous and globally well-defined.
- (ii) $\hat{f}(x, u) = f(x, u)$ for $\forall (x, u) \in X \times \mathbb{R}^p$.
- (iii) $\hat{f}(x, u)$ is globally Lipschitz in x .

Remark 2. If $\hat{f}(x, u)$ is bounded for all $(x, u) \in \mathbb{R}^n \times \mathbb{R}^p$ instead of (iii), then we call $\hat{f}$ a bounded extension.

B. Lipschitz Extension for the Observer Construction

For a given function, a Lipschitz extension can be found in general without severe difficulty. However, in our case, the extension also should preserve the lower triangular structure of $b(z, u)$. The next lemma solves the question whether there always exists an extension preserving the structure, even for the arbitrarily shaped set Z .

Lemma 5. Suppose $b(\cdot, u)$ is defined on an open set $\bar{Z} \subset \mathbb{R}^n$ and a compact set $Z \subset \bar{Z}$. Then, there exists a Lipschitz extension of $b(\cdot, u)$ from Z preserving the structure of $b(\cdot, u)$. In particular, for each element $b_i(z_1, \dots, z_i, u)$ of $b(z, u)$, there exists an extension $\hat{b}_i(z_1, \dots, z_i, u)$ such that

(i) $\hat{b}_i(\cdot, u)$ is well defined on $\mathbb{R}^i$.

(ii) $\hat{b}_i(\cdot, u) = b_i(\cdot, u)$ on the projection of Z into $\mathbb{R}^i$, i.e., on the set $\{z \in \mathbb{R}^i \mid \exists w \text{ s.t. } (z, w) \in Z\}$.

(iii) $\hat{b}_i(\cdot, u)$ is globally Lipschitz and bounded on $\mathbb{R}^i$.

Remark 3. In this Lemma, the input u is regarded as a parameter. Therefore, the properties (i) – (iii) should hold for each u with the actual bound and the Lipschitz coefficient depending on the value of u .

Proof. Let z_a and z_b be state partitions as $z_a = (z_1, \dots, z_i)$ and $z_b = (z_{i+1}, \dots, z_n)$. Define the projection of Z into $\mathbb{R}^i$ as

$$P := \{z_a \in \mathbb{R}^i \mid \exists z_b \text{ s.t. } (z_a, z_b) \in Z\} \quad (9)$$

Similarly, let Q be the projection of $\bar{Z}$ into $\mathbb{R}^i$. Then, it can be easily shown that $P \subset Q$ where P is compact and Q is open. Define a distance function $\rho(\cdot) : \mathbb{R}^i \rightarrow \mathbb{R}_{\geq 0}$ as

$$\rho(z_a) := \min_{z_p \in P} \|z_a - z_p\| \quad (10)$$

Then, it can be shown that $\rho(\cdot)$ is globally Lipschitz in $\mathbb{R}^i$ and there is $\delta > 0$ such that $P \subset \{z_a \in \mathbb{R}^i \mid \rho(z_a) \leq \delta\} \subset Q$ (see, e.g. (Munkres, 1991, Thm. 4.6)). Finally, define

$$\hat{b}_i(z_a, u) := \begin{cases} (1 - \frac{1}{\delta}\rho(z_a))b_i(z_a, u), & \text{if } 0 \leq \rho(z_a) \leq \delta \\ 0, & \text{if } \delta < \rho(z_a) \end{cases} \quad (11)$$

Then, $\hat{b}_i(\cdot, u)$ is globally well-defined and continuous. Since $\hat{b}_i(\cdot, u)$ is continuous on $\mathbb{R}^i$ and $\rho(\cdot)$ and $b_i(\cdot, u)$ are Lipschitz on the compact region $\{z_a \in \mathbb{R}^i \mid 0 \leq \rho(z_a) \leq \delta\}$, the extension $\hat{b}_i$ is globally bounded and Lipschitz by Lemma 2 and 3. $\square$

Remark 4. While the extension $\hat{b}$ in Lemma 5 is not smooth, there is even a smooth extension which satisfies global Lipschitz and bounded properties, and its existence can be proved by slightly modifying the proof of Lemma 5 and by Conlon (1992, Thm. 2.6.2). However, the proof is based on partitions of unity and is therefore nonconstructive.

Since it is not so practical to find the extension by Lemma 5, more practical extensions are implemented here when the region Z is rectangular, i.e.,

$$Z = \{z \in \mathbb{R}^n \mid -\rho_i \leq z_i \leq \rho_i, 1 \leq i \leq n\} \quad (12)$$

where ρ_i 's are certain positive constants. For simplicity, let $\bar{Z}$ be sufficiently large in the following discussions. The methods are classified according to their smoothness, assuming that the given function (e.g. $b(z, u)$) is smooth.

B.1 Smooth Lipschitz Extension

Define a smooth function $\psi(\cdot, \rho) : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\psi(s, \rho) := \begin{cases} 1, & \text{if } |s| \leq \rho \\ 1 - \frac{1}{c} \int_{\rho}^{|s|} \exp(\frac{-1}{\tau-\rho}) \exp(\frac{-1}{\rho+1-\tau}) d\tau, & \text{if } \rho < |s| < \rho+1 \\ 0, & \text{if } \rho+1 \leq |s| \end{cases}$$

where $c := \int_{\rho}^{\rho+1} \exp(\frac{-1}{\tau-\rho}) \exp(\frac{-1}{\rho+1-\tau}) d\tau$, and define the smooth function $\chi_i(\cdot) : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\chi_i(s) := \int_0^s \psi(\tau, \rho_i) d\tau.$$

Let

$$\hat{b}_i := b_i(\chi_1(z_1), \dots, \chi_i(z_i), u). \quad (13)$$

Then $\hat{b}_i$, ($1 \leq i \leq n$), becomes the smooth extension of b_i (the i -th element of $b(z, u)$). The verification is left to the reader.

B.2 C^2 Lipschitz Extension

Define $\chi_i(s) : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\chi_i(s) := \begin{cases} s, & \text{if } -\rho_i < s < \rho_i \\ \tanh(s - \rho_i) + \rho_i, & \text{if } \rho_i \leq s \\ \tanh(s + \rho_i) - \rho_i, & \text{if } s \leq -\rho_i \end{cases}$$

then, χ_i is globally bounded (i.e. $-\rho_i - 1 \leq \chi_i(s) \leq \rho_i + 1$) and C^2 . Indeed, $\chi_i(s)$ is C^∞ on $(-\rho_i, \rho_i), (\rho_i, \infty)$ and $(-\infty, -\rho_i)$. At $s = \rho_i$,

$$\frac{\partial \chi_i}{\partial s}(\rho_i+) = \frac{\partial \chi_i}{\partial s}(\rho_i-) = 1$$

$$\frac{\partial^2 \chi_i}{\partial s^2}(\rho_i+) = \frac{\partial^2 \chi_i}{\partial s^2}(\rho_i-) = 0,$$

but,

$$\frac{\partial^3 \chi_i}{\partial s^3}(\rho_i+) = -2, \quad \frac{\partial^3 \chi_i}{\partial s^3}(\rho_i-) = 0.$$

Moreover,

$$\frac{\partial \chi_i}{\partial s}(s) = \begin{cases} 1, & \text{if } -\rho_i < s < \rho_i \\ 1 - \tanh^2(s - \rho_i), & \text{if } \rho_i \leq s \\ 1 - \tanh^2(s + \rho_i), & \text{if } s \leq -\rho_i \end{cases}$$

hence, $\frac{\partial \chi_i}{\partial s}$ is globally bounded.

Now, define $\hat{b}_i$ as (14). Then $\hat{b}_i$ is globally Lipschitz in z since for all $1 \leq j \leq i$,

$$\frac{\partial \hat{b}_i}{\partial x_j}(x) = \frac{\partial b_i}{\partial \chi_j}(\chi_1(x_1), \chi_2(x_2), \dots, \chi_i(x_i), u) \frac{\partial \chi_j}{\partial x_j}(x_j)$$

which is bounded by a function of u from the boundedness of $\frac{\partial \chi_i}{\partial s}$ and χ_j 's. Moreover, it can be checked that $\hat{b}(x)$ is also C^2 .

B.3 Continuous Lipschitz Extension I

Define the saturation function $\chi_i(s) : \mathbb{R} \rightarrow \mathbb{R}$ is such that

$$\chi_i(s) := \begin{cases} s, & \text{if } -\rho_i < s < \rho_i \\ \rho_i, & \text{if } \rho_i \leq s \\ -\rho_i, & \text{if } s \leq -\rho_i \end{cases}$$

Now, define $\hat{b}_i$ as (14). Then the verification follows from the next lemma.

Lemma 6. Consider a C^1 function $f(x, y) : \mathbb{R}^p \times \mathbb{R}^q \rightarrow \mathbb{R}$ which is continuous and well-defined on $X \times \mathbb{R}^q$ where

$X = \{x \in \mathbb{R}^p \mid |x_i| \leq \rho_i, 1 \leq i \leq p\}$ with $\rho_i > 0$. Then, $f(\sigma(x), y)$ is globally well-defined, equals to $f(x, y)$ if $x \in X$, and there exists $L(y)$ such that

$$|f(\sigma(x_2), y) - f(\sigma(x_1), y)| \leq L(y) \|x_2 - x_1\|, \quad (14)$$

$$\forall x_1, x_2 \in \mathbb{R}^p, \forall y \in \mathbb{R}^q$$

where $\sigma(x)$ is a component-wise saturation function which is saturated outside X .

Proof. By the Mean-Value Theorem, there exists $z \in \mathbb{R}^p$ such that

$$f(\sigma(x_2), y) - f(\sigma(x_1), y) = \frac{\partial f}{\partial x}(z, y)(\sigma(x_2) - \sigma(x_1))$$

which implies

$$\|f(\sigma(x_2), y) - f(\sigma(x_1), y) - L(y)(\sigma(x_2) - \sigma(x_1))\|$$

where $L(y)$ is the maximum of $\|\frac{\partial f}{\partial x}(z, y)\|$ with respect to x over the compact range of saturation function. Then the claim (15) follows from the fact that $\|\sigma(x_2) - \sigma(x_1)\| \leq \|x_2 - x_1\|$. $\square$

B.4 Continuous Lipschitz Extension II

Define

$$\chi_i(s) := \begin{cases} 1 & \text{when } |s| \leq \rho_i \\ \rho_i + 1 - |s| & \text{when } \rho_i \leq |s| < \rho_i + 1 \\ 0 & \text{when } \rho_i + 1 \leq |s| \end{cases} \quad (15)$$

and let $w_i : \mathbb{R}^i \rightarrow \mathbb{R}$ as

$$w_i(s_1, \dots, s_i) = \chi_1(s_1) \chi_2(s_2) \dots \chi_i(s_i) \quad (16)$$

Then, the extension $\hat{b}_i$ is

$$\hat{b}_i := w_i(z_1, \dots, z_i) b_i(z_1, \dots, z_i, u) \quad (17)$$

Let us verify this claim utilizing the following lemma.

Lemma 7. For $1 \leq i \leq n$, the function $w_i(s_1, \dots, s_i)$ is Lipschitz on $\mathbb{R}^i$.

Proof. It can be easily shown that the function $\chi_i(s_i)$ is globally Lipschitz, i.e., there is a constant L_i such that $|\chi_i(x_i) - \chi_i(y_i)| \leq L_i |x_i - y_i|$. The rest of the proof is simple. (But not trivial. In general, the product of two Lipschitz functions are not Lipschitz.) Suppose, for example, the case when $i = 3$.

$$\begin{aligned} |w_3(x) - w_3(y)| &= |\chi_1(x_1) \chi_2(x_2) \chi_3(x_3) - \chi_1(y_1) \chi_2(y_2) \chi_3(y_3)| \\ &\leq |\chi_1(x_1) \chi_2(x_2) \chi_3(x_3) - \chi_1(x_1) \chi_2(x_2) \chi_3(y_3)| \\ &\quad + |\chi_1(x_1) \chi_2(x_2) \chi_3(y_3) - \chi_1(x_1) \chi_2(y_2) \chi_3(y_3)| \\ &\quad + |\chi_1(x_1) \chi_2(y_2) \chi_3(y_3) - \chi_1(y_1) \chi_2(y_2) \chi_3(y_3)| \\ &\leq \chi_1(x_1) \chi_2(x_2) |\chi_3(x_3) - \chi_3(y_3)| \\ &\quad + \chi_1(x_1) \chi_3(y_3) |\chi_2(x_2) - \chi_2(y_2)| \\ &\quad + \chi_2(y_2) \chi_3(y_3) |\chi_1(x_1) - \chi_1(y_1)| \\ &\leq L_3 |x_3 - y_3| + L_2 |x_2 - y_2| + L_1 |x_1 - y_1| \\ &\leq L \|x - y\| \end{aligned}$$

The other cases are similar. $\square$

Now, consider $\Omega_i = \{(z_i, \dots, z_i) \mid |z_j| \leq \rho_j + 1, 1 \leq j \leq i\}$ which is compact in $\mathbb{R}^i$. On Ω_i , Lemmas 7 and 2 yield that $\hat{b}_i$ is Lipschitz in z . On its complement Ω_i^c , $\hat{b}_i$ is also Lipschitz in z since $\hat{b}_i \equiv 0$. Then, Lemma 3 leads to that $\hat{b}_i$ is globally Lipschitz in z . Therefore, $\hat{b} = [w_1 b_1, \dots, w_n b_n]^T$ gives the desired extension. Note that Ω_i is the projection of $\{z \in \mathbb{R}^n \mid |z_j| \leq \rho_j + 1, 1 \leq j \leq n\}$ into $\mathbb{R}^i$.

Remark 5. All the constructions of the Lipschitz extension in this section also give the bounded extensions.

IV. X-COORDINATE OBSERVER

Even though a system is given in x -coordinates as

$$\dot{x} = f(x) + g(x, u), \quad y = h(x), \quad (18)$$

the nonlinear observer for this system is usually designed in a special coordinates like the observer canonical form or the observable canonical form. Thus, the nonlinear coordinate transformation $z = \Phi(x)$ must be invertible. However, the calculation of the inverse $\Phi^{-1}(z)$ can be avoided for the observer designs. Suppose (19) is transformed to $\dot{z} = Az + \gamma(y, u)$ and $y = Cz$ and the observer is designed as

$$\dot{\hat{z}} = A\hat{z} + \gamma(y, u) - L(C\hat{z} - y).$$

Then, the observer in x -coordinates is obtained by

$$\dot{\hat{x}} = f(\hat{x}) + g(\hat{x}, u) - \left[\frac{\partial \Phi}{\partial \hat{x}}(\hat{x}) \right]^{-1} L(h(\hat{x}) - y)$$

where $\left[\frac{\partial \Phi}{\partial \hat{x}} \right]^{-1}$ is easily calculated by a computer, and thus, there is no need to have $\Phi^{-1}(z)$.

However, in our case of Lipschitz extension, the extension is performed in z -coordinates and therefore the dynamics is modified with respect to the state z rather than x . Hence, the analytic calculation of $\Phi^{-1}(z)$ is necessary in general. Fortunately, if the extension method (d) of the previous discussions is used, we can construct x -coordinate observer form without the knowledge of $\Phi^{-1}(z)$.

The key to success is the modification by multiplication. We provide the x -coordinate observer for (19) without further proof:

$$\dot{\hat{x}} = W_f(\hat{x})f(\hat{x}) + W_g(\hat{x})g(\hat{x}, u) - \left[\frac{\partial \Phi}{\partial \hat{x}}(\hat{x}) \right]^{-1} G^{-1} C^T (Ch(\hat{x}) - y) \quad (19)$$

where G is the solution of (17) for sufficiently large θ and

$$\begin{aligned} W_f(x) &= \left[\frac{\partial \Phi}{\partial x}(x) \right]^{-1} \cdot \text{diag}[1, \dots, 1, w_0(\Phi(x))] \cdot \left[\frac{\partial \Phi}{\partial x}(x) \right] \\ W_g(x) &= \left[\frac{\partial \Phi}{\partial x}(x) \right]^{-1} \cdot \text{diag}[w_1(\Phi_1(x)), w_2(\Phi_1(x), \Phi_2(x)), \dots, \\ &\quad \dots, w_n(\Phi_1(x), \dots, \Phi_n(x))] \cdot \left[\frac{\partial \Phi}{\partial x}(x) \right] \end{aligned}$$

where Φ_i implies the i -th element of Φ .

V. CONCLUSION

In this paper, we discussed the Gauthier's observer and pointed out the restriction of global lipschitz condition. Presenting comprehensive discussions on lipschitz extension technique, we can relieve the global lipschitz requirement. Finally, using specific extension technique in rectangular region, an x -coordinate observer is proposed.

REFERENCES

- A.J.Krener and A. Isidori (1983). Linearization by output injection and nonlinear observers. *Systems & Control Letters* **3**, 47–52.
- A.J.Krener and W. Respondek (1985). Nonlinear observers with linearizable error dynamics. *SIAM J. Control Optim.* **23**(2), 197–216.
- Besancon, G. and H. Hammouri (1996). On uniform observation of nonuniformly observable systems. *Systems & Control Letters* **29**, 9–19.
- Bestle, D. and M. Zeitz (1983). Canonical form observer for non-linear time-variable systems. *Int. J. Control* **38**, 419–431.
- Birk, J. and M. Zeitz (1988). Extended luenberger observer for non-linear multivariable systems. *Int. J. Control* **47**(6), 1823–1836.
- Conlon, L. (1992). *Differentiable Manifolds*. Birkhauser.
- Dawson, D.M., Z. Qu and J.C. Carroll (1992). On the state observation and output feedback problems for nonlinear uncertain dynamic systems. *Systems & Control Letters* **18**, 217–222.
- Gauthier, J.P. and G.Bornard (1981). Observability for any $u(t)$ of a class of nonlinear systems. *IEEE Trans. Automat. Contr.* **26**, 922–926.
- Gauthier, J.P. and I. Kupka (1994). Observability and observers for nonlinear systems. *SIAM J. Control Optim.* **32**(4), 975–994.
- Gauthier, J.P., H.Hammouri and S.Othman (1992). A simple observer for nonlinear systems: Applications to bioreactors. *IEEE Trans. Automat. Contr.* **37**(6), 875–880.
- Hou, M. and A.C. Pugh (1999). Observer with linear error dynamics for nonlinear multi-output systems. *Systems & Control Letters* **37**, 1–9.
- Joo, S.J., J. Byun, H. Shim and J.H. Seo (1994). Design and analysis of the nonlinear feedback linearizing controller for an ems system. In: *3rd IEEE Conf. on Control Applications*. pp. 593–598.
- Keller, H. (1987). Non-linear observer design by transformation into a generalized observer canonical form. *Int. J. Control* **46**(6), 1915–1930.
- Li, C.W. and L.W. Tao (1986). Observing non-linear time-variable systems through a canonical form observer. *Int. J. Control* **44**(6), 1703–1713.
- Munkres, J.R. (1991). *Analysis on Manifolds*. Addison Wesley.
- Nijmeijer, H. and A.J. van der Schaft (1990). *Nonlinear Dynamical Control Systems*. Springer-Verlag.
- Proychev, T.P. and R.L. Mishkov (1993). Transformation of nonlinear systems in observer canonical form with reduced dependency on derivatives of the input. *Automatica* **29**(2), 495–498.
- Rajamani, R. and Y. Cho (1995). Observer design for nonlinear systems: stability and convergence. In: *Conf. on Decision and Control*. pp. 93–94.
- Rayamani, R. (1998). Observers for lipschitz nonlinear systems. *IEEE Trans. Automat. Contr.* **43**(3), 397–401.
- Shim, H., Y.I. Son and Jin H. Seo (1999). Saturation technique for constructing observer of multi-output nonlinear systems. pp. 3077–3081.
- Tsinias, J. (1989). Observer design for nonlinear systems. *Systems & Control Letters* **13**, 135–142.
- Vidyasagar, M. (1993). *Nonlinear systems analysis*. Prentice-Hall.
- Walcott, B.L. and S.H. Zak (1987). State observation of nonlinear uncertain dynamical systems. *IEEE Trans. Automat. Contr.* **32**(2), 166–170.
- Xia, X.H. and W.B. Gao (1988). Non-linear observer design by observer canonical form. *Int. J. Control* **47**(4), 1081–1100.
- Xia, X.H. and W.B. Gao (1989). Nonlinear observer design by observer error linearization. *SIAM J. Control Optim.* **27**(1), 199–216.